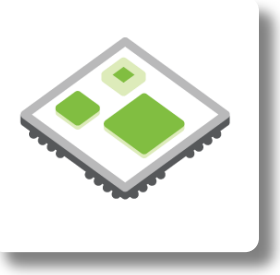
SERVER:
OPEN CHIPLET
ECONOMY

Security of Multi-chip ICs & Security Chiplets with Caliptra: Building a Hardware Root of Trust



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Chiplet Security requirements

- Chiplet identity and provenance
- Secure Boot and measured boot
- Inter-die communication channel encryption
- Lifecycle governance and update management
- Provisioning and key management
- Telemetry
- Traceability
- Die-to-die links carry traffic that may cross organizational boundaries
- IP Protection

Secure boot, measured boot, and root of trust

- Establish a hardware root of trust that verifies chiplet firmware, configuration, and interconnect components.
- Caliptra measured boot: A system that can prove which chiplets and software were loaded. Records and attests cryptographic hashes of system firmware and configuration data into hardware registers before execution to establish a verifiable chain of trust
- Define a consistent boot-time trust sequence across vendors.
- Use physical techniques such as Fuse Shadow Registers. Die Identification: Unique serial numbers, lot codes, revision ID’s, etc.
- Caliptra Unique Device Secret (UDS) establishes silicon root of trust.

Fuse Shadow Registers

- Individual Dies use their own Efuses for configuration, collaboration, ID coding and security keys.
- Die Identification: Unique serial numbers, lot codes, revision ID’s, etc.
- Lifecycle and security flags.
- Anti-Rollback: Locking software version numbers to prevent downgrading firmware.

Inter-die communication channel encryption

- Protect confidentiality, integrity, freshness, and availability of traffic across the die-to-die link.
- Define secure behavior for link initialization, reset, power transitions, retries, and error handling.

Interface and protocol security

- Specify security requirements for protocols such as **UCIe** and implementations based on **BoW** and **BoW.flexi**
- UCIe provides a standardized die-to-die ecosystem, but system designers must configure and validate its security capabilities appropriately.
- BoW primarily defines an open physical die-to-die interface; encryption, authentication, access control, and lifecycle security generally require additional implementation-specific mechanisms.

Caliptra and DICE (Device ID Composition Engine)

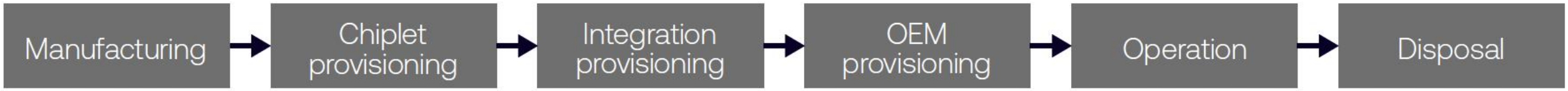
- Starts with a Unique Device Secret (UDS)
- **Layered Derivation:** Combines the secret with measurements of the next boot component to create a chain of trust
- Caliptra supplies hardware-rooted security foundation
- DICE defines a composable identity and measurement model
- Used together, they allow each chiplet to prove not only *who it is*, but also *what firmware and security state it is currently running*.

Caliptra secure provisioning and key management

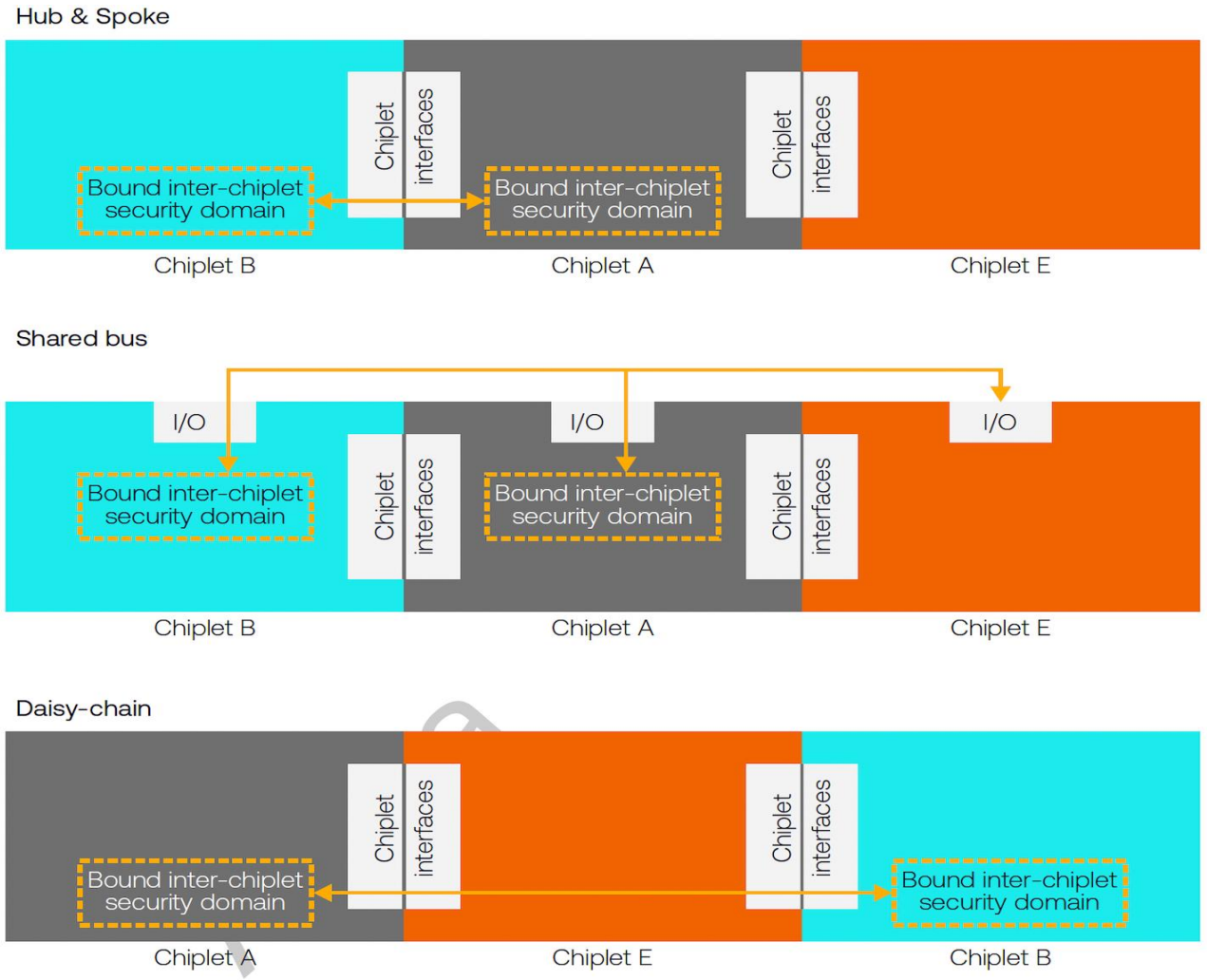
- Caliptra provides an open-source silicon root of trust that delivers hardware-enforced secure provisioning, key management, and cryptographic services
- Protect confidentiality, integrity, freshness, and availability of traffic across the die-to-die link.
- Define secure behavior for link initialization, reset, power transitions, retries, and error handling.

Lifecycle governance

- The systematic process of tracking, securing, testing, and governing individual chiplets from initial design and multi-vendor sourcing through manufacturing, packaging, and in-field deployment.



Chiplet topology models



Two Chiplet Boot scenario

- **Primary Chiplet (Orchestrator):** Manages the global system boot policy, aggregates overall system health status, and interfaces with the outside world. Hardware Root-of-Trust
- **Secondary Chiplet (Subordinate):** Operates under its own local security boundary, executing tasks (like matrix multiplication or network IO) only after validating itself to the Primary.
- **Mutual Authentication**: After basic channels are setup, establish a distributed chain of trust. The main SoC Root of Trust (RoT) and the individual chiplet sub-RoTs must cryptographically prove their identity and firmware integrity to one another before continuing.

Key Takeaways

- Chiplets expand the attack surface across design, package, interconnect, and lifecycle
- Interposer probing is a credible physical threat to confidentiality and integrity
- UCIe and BoW enable connectivity but do not replace system-level security architecture. UCIe 3.0 adds die to die security and management features
- Identity, authorization, protected transport, monitoring, and recovery work together
- Security must be verified as an end-to-end package property
- Beware of FPGA security issues! An exposed system can be attacked.

Call to Action

- Problem to solve
  - Establish more support for evolving standards
  - Support methods of backwards compatibility
- OCP Open Chiplet Economy is in the Server project, and Caliptra is in the Security project. Participation is open and easy.
- OCP Foundation Chiplet System Architecture (FCSA) version 1.1 Chapter 9!
- OCP Academy - Open Chiplet Economy: Business and Technology of Chiplet-based Systems - 9-course series in chiplet systems. Free of charge!

Thank You!

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